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Monetizing achievement: flow experience, game genre, and in-game consumption in the marketing communication of virtual goods

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Background and objective: Online games combine gameplay with commercial communication through rankings, rewards, item descriptions, progression systems, and competitive feedback. These features communicate the functional and symbolic value of virtual goods, yet the psychological processes associated with players' consumption responses remain insufficiently explained within Advertising and Marketing Communication. This study examines the associations among Achievement Motivation, Flow Experience, Game Genre, Player Type, and Online Game Consumption Behavior by integrating achievement motivation theory, flow theory, and person–situation interactionism.

Methods: A laboratory-based cross-sectional correlational study was conducted with 300 online game players recruited through Credamo in 2024. Credamo was used only for participant recruitment. Participants attended an offline laboratory in Hengqin, Guangdong, played their preferred online game for 60 min, and completed a post-session questionnaire. Online Game Consumption Behavior was represented by four ordered categories of monthly in-game expenditure. Player Type was classified as light or heavy according to weekly gaming time, while Game Genre was coded as a competitive or noncompetitive game context. The proposed associations were examined using regression-based moderation and mediation analyses with 5,000 bootstrap samples.

Results: Achievement Motivation was positively associated with Online Game Consumption Behavior (B = 0.609, $p < .001$) and Flow Experience (B = 0.528, $p < .001$). Flow Experience remained positively associated with Online Game Consumption Behavior after Achievement Motivation was included in the model (B = 0.617, $p < .001$). The indirect association through Flow Experience was statistically different from zero [B = 0.326, 95% CI (0.229, 0.479)], while the direct association remained significant (B = 0.283, $p < .001$). Player Type did not significantly moderate the association between Achievement Motivation and Online Game Consumption Behavior (B = –0.044, $p = .173$). Game Genre showed a significant moderating association (B = 0.220, $p < .001$), with a stronger positive association in competitive game contexts.

Conclusions: The findings connect online game consumption with a commercial communication environment formed through gameplay systems and purchase-related information. Flow Experience statistically accounted for part of the association between Achievement Motivation and Online Game Consumption Behavior, while Game Genre was associated with variation in the strength of this relationship, with a stronger positive association in genres coded as competitive. The results suggest that the meaning communicated through virtual goods may be more closely related to player motivation when competition and visible performance comparison are central to gameplay.

KEYWORDS

achievement motivation, digital marketing communication, embedded commercial communication, flow experience, game genre, in-game spending, virtual goods

1 Introduction

Digital games have developed into platform-based commercial environments in which entertainment, social interaction, and consumption are closely intertwined. In-game purchases—including functional items, cosmetic goods, character upgrades, and access to additional content—have become an important source of revenue within this environment. These virtual goods may communicate functional, hedonic, social, and symbolic value to players (Costes and Bonnaire, 2022; Hamari, 2015; Hamari and Keronen, 2017; Song, 2024). Their value is conveyed not only through conventional promotional messages but also through item descriptions, rarity classifications, reward systems, progress indicators, rankings, and purchase interfaces embedded within gameplay. Commercial communication in digital games is therefore frequently integrated into the experience rather than presented as a separate interruption.

Advertising research has traditionally examined digital games as channels for brand placement, in-game advertising, and advergames (Nelson, 2002; Terlutter and Capella, 2013). This perspective has established the relevance of games as persuasive media, particularly by demonstrating how interactive environments shape attention, brand memory, and responses to commercial stimuli. Contemporary game monetization extends this relationship further. The platform itself continually communicates which virtual goods are valuable, how they relate to progression, and what forms of performance or status they make attainable. Game mechanics thus operate alongside explicit promotional content as part of the commercial communication environment. Research on gamification similarly indicates that goals, feedback, rewards, and progression systems can structure consumer engagement with marketing platforms (Hofacker et al., 2016). Examining in-game spending from this perspective shifts attention from whether players encounter a commercial message to how they interpret monetized opportunities embedded within play.

Players are unlikely to interpret these opportunities uniformly. Achievement-oriented players place greater importance on mastery, performance improvement, challenging goals, and visible indicators of success. Rankings, level systems, performance feedback, and competitively useful virtual goods may consequently hold greater relevance for these players than for those who approach a game primarily for relaxation or social interaction. Achievement motivation theory provides an appropriate foundation for explaining this variation. It suggests that individuals differ in the value they assign to challenging tasks and demonstrable accomplishment (Atkinson, 1957; McClelland, 1961). In a digital game, virtual goods can become associated with achievement when they are presented as supporting progression, improving performance, or making competence and status visible.

The association between achievement motivation and spending may also be understood through the quality of the gameplay experience. Flow refers to a state of concentrated absorption characterized by focused attention, clear goals, immediate feedback, and a perceived balance between challenge and skill (Cai et al., 2022; Csikszentmihalyi, 1990). Achievement-oriented players may report stronger flow experiences when the game offers meaningful challenges and continuous performance feedback. Flow, in turn, may be associated with spending when virtual goods are presented as relevant to continued progression, preferred forms of play, or access to valued gameplay experiences. This interpretation does not equate enjoyment with purchasing. Research on virtual goods indicates that game enjoyment, continued use, attitudes toward virtual purchases, and spending may follow distinct patterns (Hamari, 2015).

The relevance of achievement cues also depends on the context in which they are encountered. Competitive game environments organize play around direct comparison, visible rankings, performance differentials, and recurring feedback about winning and losing. These features can make the achievement-related meaning of virtual goods more salient. In less competitive environments, progression and enjoyment may remain important, but purchasing opportunities are less consistently interpreted through direct comparison with other players. A person–situation interaction perspective therefore suggests that the association between achievement motivation and spending may differ across competitive and noncompetitive game contexts. Weekly playing time may provide an additional boundary condition. Players who spend more time in games typically encounter monetization systems more frequently and possess greater familiarity with the functional and symbolic meanings assigned to virtual goods.

The proposed model is tested using data from 300 online game players in China who completed a standardized gameplay session in a controlled offline laboratory and responded to an immediate post-session survey. The research is characterized as a laboratory-based cross-sectional correlational study. Achievement Motivation, Flow Experience, and the contextual variables were measured using self-reported responses, while Online Game Consumption Behavior was represented by four ordered categories of self-

reported monthly in-game expenditure during the preceding 30 days. The analyses evaluate a moderated mediation model in which Flow Experience statistically accounts for part of the association between Achievement Motivation and the ordered expenditure score, with Game Genre and Player Type examined as conditional factors.

This study contributes to Advertising and Marketing Communication by conceptualizing gameplay and monetization systems as part of the environment through which the value of virtual goods is communicated. Instead of treating commercial persuasion only as an external advertisement placed within a game, the study considers how rankings, rewards, progression systems, and competitive feedback frame paid items as resources for achievement. It also connects this communication perspective with achievement motivation and flow theory, providing an integrated account of motivational relevance, gameplay experience, and self-reported consumer expenditure. The moderated model further indicates that responses to monetized game environments vary across game genres and patterns of player participation. For practitioners, the findings may support clearer and more context-sensitive communication surrounding virtual goods, while also emphasizing transparent monetization and the responsible treatment of achievement-oriented players.

2 Literature review

2.1 Achievement motivation in gaming and virtual consumption

Achievement motivation explains individual differences in the pursuit of competence, mastery, and performance standards. Atkinson (1957), McClelland et al. (1953), and McClelland (1961) describe achievement-oriented behavior as involving the selection of challenging tasks, sustained effort, attention to performance feedback, and evaluation against standards of success (Yilmaz and Tunca, 2023). These features make achievement motivation especially relevant to digital games, where progress is organized through explicit goals, levels, rankings, rewards and repeated evaluations of player performance (Zhao et al., 2022).

Yee (2006) identifies achievement as a major motivation for gameplay, alongside social interaction and immersion (Holl et al., 2024; Wang et al., 2021). Achievement-oriented players tend to place greater value on advancement, skill development, accumulated resources, and visible indications of competence (Brühlmann et al., 2020; Hygen et al., 2024; Kępka and Strzelecki, 2024; Wang and Cheng, 2022; Zhou et al., 2022). Their engagement with a game is therefore connected to how clearly the game communicates progress and how strongly its reward structure recognizes successful performance. Levels, badges, leaderboards, equipment scores, and competitive rankings provide information about current performance while also defining what counts as success within the game.

Virtual goods form part of this achievement system when their functions or symbolic meanings are connected to progress. Functional items may support task completion, character development, or access to additional content. Cosmetic and status-related items may communicate experience, rarity, group membership, or distinction among players. Costes and Bonnaire (2022), Hamari and Keronen (2017), Lelonek-Kuleta et al. (2021), and Lemmens (2022) associate virtual-item purchasing with achievement-related motives, social value, and perceived utility (Tuzzahra and Edastama, 2023). These findings indicate that an in-game purchase cannot be understood only as payment for a digital object. The purchase also reflects how the player interprets the object within the game's system of goals and rewards.

Virtual goods are presented through descriptions, performance statistics, visual rarity cues, limited availability, reward comparisons, and placement within the game interface. These cues communicate what the item offers and how it may contribute to the player's desired position in the game. Achievement-oriented players may assign greater value to purchase opportunities that are connected to mastery, progress, or visible performance. The same virtual item may therefore carry different levels of relevance across players, even when its price and technical features remain unchanged.

Achievement motivation provides a theoretical explanation for this variation in perceived relevance. It does not imply that all achievement-oriented players will spend money or that each purchase is motivated by performance. In-game spending can also reflect enjoyment, customization, social interaction, identity expression, convenience, and other player-, game-, platform-, and contextual factors (Hamari, 2015; Song, 2024; Zhang, 2024). The present argument is more specific: when a game communicates virtual goods in achievement-related terms, players with stronger achievement motivation are expected to report higher spending than players for whom mastery and performance hold less personal value.

2.2 Flow experience in digital games

Flow describes a state of focused involvement in which attention is concentrated on an activity and the person perceives a suitable balance between challenge and skill (Csikszentmihalyi, 1990; Keller and Bless, 2008). Clear goals, immediate feedback, concentration, a sense of control, and reduced awareness of time are commonly associated with this experience. Digital games frequently provide these conditions through mission objectives, difficulty systems, performance indicators, and rapid feedback on player actions (Cai et al., 2022; Footitt et al., 2024; Sweetser and Wyeth, 2005).

Achievement-oriented individuals tend to value tasks that provide meaningful standards of performance and opportunities to improve their competence. Tasks that are too simple provide little basis for accomplishment, while tasks perceived as impossible may reduce continued effort. Games that present demanding but manageable objectives offer conditions under which achievement-oriented players may report stronger concentration and involvement (Hong et al., 2020; Lin et al., 2025; Xu, 2024). The association between achievement motivation and flow therefore rests on a shared emphasis on challenge, skill, goals, and performance feedback.

Flow has commonly been connected to satisfaction, continued use, loyalty, and positive evaluations of digital services (Choi and Kim, 2004; Finneran and Zhang, 2003, 2005; Hsu and Lu, 2004). Its relationship with purchasing requires a more careful interpretation. Enjoyment and spending are not equivalent outcomes. Hamari (2015), for example, reports that game enjoyment may reduce the perceived need to purchase virtual goods while continued use and favorable attitudes toward virtual purchases remain positively related to purchasing intentions.

Flow may consequently be associated with spending under specific commercial conditions, especially when a virtual item is communicated as relevant to progress, preferred gameplay, or access to valued content.

A player experiencing strong involvement may assign greater value to an item that supports a personally important game objective. This does not mean that the purchase creates or preserves flow in every case. Some purchases are cosmetic, some respond to social expectations, and others reduce inconvenience. The theoretical connection concerns the value attached to continued participation in the game. When gameplay is experienced as focused and rewarding, goods associated with that experience may acquire greater functional or symbolic value. Chou et al. (2023) similarly connect flow-related gameplay experiences with in-game purchase responses across gaming platforms.

Within the present theoretical model, flow describes the experiential connection between achievement motivation and spending. Achievement motivation concerns the importance assigned to mastery and performance, while flow concerns the quality of involvement experienced during gameplay. In-game spending represents a separate economic response to the commercial opportunities presented by the platform. Treating these constructs separately prevents the analysis from assuming that motivation, enjoyment, involvement, and purchasing are interchangeable forms of player engagement.

2.3 Person–situation interaction and game genre

Person–situation interactionism explains behavior through the joint relevance of individual characteristics and situational conditions. Lewin (1936) expressed this position as $(B = f(P, S)$, where behavior reflects the person and the situation in which the behavior occurs. Mischel (1977) further argued that the expression of individual characteristics varies across situations with different rules, expectations, and behavioral cues. Applied to digital games, achievement motivation represents a personal characteristic, while the organization of goals, competition, feedback, and social comparison forms part of the situation.

Game genres differ in the skills they require, the goals they emphasize, and the forms of interaction they organize (Brandtner et al., 2022). In the present study, Game Genre is treated as a situational variable and coded according to the dominant competitive structure of gameplay. Genres organized around direct opposition, visible performance comparison, rankings, and outcomes defined through winning and losing were classified as competitive. Genres in which competition was secondary to narrative progression, exploration, collection, creation, or individual problem solving were classified as noncompetitive. This distinction concerns the dominant organization of gameplay and does not suggest that every title within a genre offers an identical experience.

Competitive game contexts may increase the relevance of achievement motivation by making differences in performance visible. Rankings and competitive feedback communicate the player's position relative to others, while reward structures identify the behaviors and outcomes valued by the game. Virtual goods presented within this setting may also receive a stronger achievement-related meaning when they are associated with performance, progression, distinction, or access to competitive content. Jang et al. (2021) and Ma et al. (2021) show that genre-related characteristics can change the relationships between player characteristics and gaming responses. The association between achievement motivation and spending may therefore be stronger when the dominant game context makes achievement public, comparable, and repeatedly evaluated.

Player type represents a second source of variation in this relationship. In this study, player type is based on weekly playing time and distinguishes heavy players from light players. This measure reflects the extent of participation during a typical week; it is not treated as a personality trait or as a direct measure of consumer involvement. Heavy players generally have more frequent contact with game systems, virtual goods, pricing information, and promotional prompts. They may also possess greater knowledge of the functional and symbolic meanings assigned to particular items. Light players encounter the same commercial environment less frequently and may assign lower personal importance to game-related performance.

Weekly playing time may therefore change the strength of the association between achievement motivation and spending. Greater exposure can increase familiarity with purchase options and provide more occasions for achievement-related motives to become relevant. At the same time, playtime alone does not establish stronger psychological involvement or a greater willingness to pay. Heavy players may learn to progress without making purchases, while light players may make occasional high-value purchases.

Achievement Motivation, Flow Experience, Game Genre, and Player Type describe different parts of the proposed explanation. Achievement Motivation concerns the value players assign to mastery and successful performance. Flow Experience concerns their reported concentration and involvement during gameplay. Game Genre captures the dominant competitive structure through which performance is organized and communicated. Player Type captures the frequency of participation and exposure to the game environment.

2.4 Online games in diverse environments

Online games comprise a broad set of digital environments that differ in their technological platforms, rule systems, player activities, social structures, and commercial models. They can be understood as sociotechnical systems in which players interact with game content and with other players under a defined set of rules. These environments support several forms of participation, including competition, cooperation, exploration, narrative progression, collection, and character development. Their structural diversity is relevant to the study of player behavior, as the meaning assigned to achievement and virtual goods may vary across different forms of gameplay (Liao et al., 2023; Rehbein et al., 2021; Zhang et al., 2021).

Online games can first be classified according to the device through which they are accessed. Home consoles, handheld consoles, arcade machines, personal computers, and mobile devices differ in accessibility, interface design, usage setting, and typical patterns of participation. Device classification provides the technological context in which gameplay and purchasing

occur. It does not, however, fully describe the goals, challenges, or forms of interaction offered by a game. Genre classification provides additional information about the activities that organize the player's experience.

Game genres are commonly distinguished according to their central mechanics, goals, and required player skills, and genre differences may also correspond with different gaming motivations (Brandtner et al., 2022; Wohn et al., 2020). Role-playing games emphasize character development, narrative progression, and resource acquisition. Strategy games focus on planning, resource allocation, and tactical decisions. Shooting, fighting, racing, and sports games place greater emphasis on direct performance and competition. Puzzle and simulation games commonly organize play around problem solving, creation, or individual progression. These differences influence the types of achievement that players pursue and the value they may assign to game-related rewards.

Table 1 presents five major game-device categories and twelve principal game genres. The table is not itself the binary moderator used in the statistical analysis. The empirical analysis classifies the genres represented in the sample according to the dominant competitive structure of gameplay.

Competitive structure was selected as the focal genre characteristic due to its connection with achievement motivation and commercial communication. In games organized around direct player competition, rankings, performance statistics, and win–loss outcomes make achievement visible and comparable. In games where direct competition is secondary, achievement may be expressed through exploration, narrative completion, collection, character development, or individual problem solving. The broader genre classification shown in Table 1 therefore establishes the diversity of online games, while the competitive-context classification supports the moderator examined in the proposed model.

3 Hypothesis and research model

3.1 Achievement motivation and consumption behavior

Achievement motivation reflects the importance that individuals assign to mastery, competence, and successful performance. Within online games, these goals are pursued through level advancement, task completion, skill improvement, rankings, and competitive outcomes (Wang and Cheng, 2022; Zhou et al., 2022). Players with stronger achievement motivation are therefore expected to pay closer attention to game features and virtual goods associated with progress or performance.

In-game purchase opportunities commonly communicate the functional and symbolic value of virtual goods through item attributes, performance statistics, rarity cues, reward comparisons, and progress-related messages. Some virtual goods provide functional assistance, such as improved equipment, additional abilities, or access to content. Other goods communicate status, experience, or distinction among players (Costes and Bonnaire, 2022; Hamari and Keronen, 2017). The relevance assigned to these offers is likely to vary according to the player's goals. A performance-related item may hold greater value for a player strongly motivated by advancement than for a player whose participation is mainly social or recreational.

Achievement motivation does not imply that players will necessarily purchase virtual goods, as spending is also related to price, disposable income, game design, social influence, and attitudes toward virtual purchases. The proposed relationship concerns the greater relevance that achievement-oriented players may assign to goods presented as supporting progress, competence, or visible accomplishment. This interpretation connects the player's motivation with the performance-related value communicated by the game system. We therefore propose the following hypothesis:

H1: Achievement motivation is positively associated with online game consumption behavior.

3.2 Role of player type

Player type in this study is defined according to weekly playing time, distinguishing heavy players from light players Hamutoğlu et al., 2020. Weekly playing time indicates the frequency and duration of participation in digital games. It does not directly measure psychological involvement, player identity, or emotional attachment. Heavy players generally have more frequent contact with gameplay systems, virtual-item stores, price information, promotional messages, and progress-related purchase opportunities. This repeated contact may increase their familiarity with the functions and symbolic meanings of virtual goods.

Achievement-oriented heavy players may also encounter more occasions in which game goals, performance feedback, and purchase opportunities appear together. Their knowledge of game mechanics may allow them to evaluate more clearly whether a paid item contributes to progress or performance. Among light players, achievement motivation can still be relevant, but less frequent participation provides fewer opportunities for achievement-related goals to become connected with spending decisions.

Weekly playing time alone does not guarantee greater expenditure. Heavy players may develop sufficient skills to progress without purchases, while light players may make occasional high-value purchases. The proposed moderation therefore concerns the strength of the association between achievement motivation and spending, not a general difference in spending between heavy and light players. Accordingly:

H2: Player type moderates the positive relationship between achievement motivation and online game consumption behavior, such that the relationship is stronger for heavy players than for light players.

3.3 Role of game genre

Game genres differ in their central mechanics, goals, required skills, and forms of player interaction (Brandtner et al., 2022; Liao et al., 2023). For the present model, game genre is considered according to the dominant competitive context of gameplay. Competitive genres, including multiplayer online battle arena and first-person or third-person shooting games, commonly organize play around direct opposition, visible rankings, win–loss records, and repeated comparison between players. In noncompetitive genres, competition may be optional or

TABLE 1 Five major game device categories and twelve principal game genres.

Type	Name	Description	Representative Games/ Devices
Category	Home Console	Home-oriented gaming devices, connected to a television or monitor and operated with a controller.	HOME-PONG, Famicom, SG-1000
Category	Handheld Console	Small portable gaming devices that allow play anywhere and anytime.	Nomad
Category	Arcade	Specialized gaming machines installed in public entertainment venues.	The King of Fighters, Contra, Street Fighter
Category	Computer Games	Games developed for personal computers, covering a broad set of genres as technology advances.	See genre rows below
Category	Mobile Games	Games that run on mobile telephones.	Honor of Kings (Arena of Valor)
Genre	Action Game (ACT)	Focuses on players' reaction speed and agility.	Devil May Cry
Genre	Adventure Game (AVG)	Players solve tasks with items and clues while following a narrative.	Sekiro: Shadows Die Twice
Genre	Role Playing Game (RPG)	Players assume character roles, interact with others, complete tasks, and develop skills and equipment.	The Witcher
Genre	Strategy Game (STR)	Centers on strategic use of resources, exploration, and technological development.	This War of Mine
Genre	Simulation Game (SLG)	Replicates real-world environments, movements, or activities for immersive play.	The Sims
Genre	Sports Game (SPG)	Simulates sports competitions, allowing play as athletes, managers, or coaches.	NBA 2K21
Genre	Puzzle Game (PZG)	Challenges logic and reasoning; requires careful thinking during play.	Minesweeper
Genre	Shooting Game (STG)	Players use weapons to eliminate enemies, avoid attacks, and win.	Counter-Strike
Genre	Race Game (RCG)	Emphasizes vehicle control and concentration in races.	CrazyRacing KartRider
Genre	Fighting Game (FGT)	Players control characters in one-on-one matches to defeat opponents.	Street Fighter
Genre	Educational Game (EDU)	Games created primarily for education.	Poly Bridge
Genre	Training/Development Game (TDG)	Focuses on nurturing an object or character, often with management elements.	Miracle Nikki

The table summarizes the broader device and genre classification used to describe online game environments. The binary coding of competitive and noncompetitive game contexts used in the statistical analysis is reported in the Research Method section.

secondary to activities such as narrative progression, exploration, collection, or individual problem solving.

Competitive game contexts communicate achievement through public and measurable performance information. Ranking systems, competitive ladders, match results, and performance statistics make differences between players visible (Hattingh et al., 2024; Jang et al., 2021; Ma et al., 2021). These features may increase the relevance of achievement motivation by connecting personal goals with repeated social comparison. They may also strengthen the perceived value of virtual goods presented as supporting performance, progression, or competitive distinction.

In less competitive contexts, achievement can remain personally important, but it is less consistently evaluated through direct comparison with other players. Virtual goods may consequently be interpreted in terms of customization, exploration, convenience, or narrative progress, with a weaker connection between achievement motivation and expenditure. The argument concerns the dominant competitive structure of the genre and does not assume that all games within a genre provide identical mechanics or monetization systems. Thus:

H3: Game genre moderates the positive relationship between achievement motivation and online game consumption behavior, such that the relationship is stronger in competitive game contexts than in noncompetitive game contexts.

3.4 Process of flow experience

Flow experience provides the experiential component of the proposed model. Flow is characterized by concentrated attention, a perceived balance between challenge and skill, clear goals, immediate feedback, and a sense of control during an activity (Cai et al., 2022; Chang et al., 2020). These conditions correspond closely with the goals valued by achievement-oriented players. Players with stronger achievement motivation tend to seek challenging tasks, monitor their performance, and persist in improving their skills (Lin et al., 2025; Wibowo and Hardiwinata, 2023). When game demands are perceived as challenging but manageable, this orientation may be associated with stronger flow experience.

The proposed relationship does not assume that achievement motivation always produces flow. Flow also depends on task difficulty, player skill, feedback quality, interruptions, and the design of the gameplay session. Achievement motivation is expected to increase the player's attention to challenging goals and performance feedback, which supports a positive association with reported flow. Therefore:

H4a: Achievement motivation is positively associated with flow experience.

Flow may also be associated with in-game spending. A highly focused and rewarding gameplay experience can increase the personal value assigned to continued participation and to the

goods offered within that environment. Players reporting stronger flow may evaluate certain virtual items more favorably when those items are connected with desired content, progress, or preferred forms of gameplay (Chou et al., 2023; Song, 2024).

This relationship should not be interpreted as an assumption that players purchase items to maintain flow or that flow necessarily results in expenditure. Virtual goods serve different functions, and some purchases have little connection with immediate gameplay performance. The theoretical expectation is limited to a positive association: players who report stronger flow may also assign greater value to purchase opportunities associated with an experience they consider rewarding. Accordingly:

H4b: Flow experience is positively associated with online game consumption behavior.

The relationships proposed in H4a and H4b support an indirect association between achievement motivation and in-game spending. Achievement-oriented players may report stronger flow during gameplay, and stronger flow may be accompanied by higher reported spending. Flow therefore accounts for part of the statistical association between achievement motivation and expenditure, while a direct association may remain after flow is included in the model. Thus:

H4: Flow experience mediates the relationship between achievement motivation and online game consumption behavior.

The hypotheses are summarized in the conceptual research model shown in Figure 1. H1 proposes a positive association between Achievement Motivation and Online Game Consumption Behavior. H2 and H3 propose that this association varies according to Player Type and Game Genre, respectively. H4 proposes an indirect association through Flow Experience, comprising the positive association between Achievement Motivation and Flow Experience identified in H4a and the positive association between Flow Experience and Online Game Consumption Behavior identified in H4b.

4 Research method

4.1 Measures, sample and data collection

This study used a controlled, laboratory-based cross-sectional correlational design to examine the relationships among players' achievement motivation, flow experience, player type, game genre, and online game consumption behavior. No variable was experimentally manipulated, and participants were not randomly assigned to study conditions. The laboratory procedure provided consistent gameplay duration and data-collection conditions, while the focal variables were measured through an immediate post-session questionnaire and factual questions about participants' game consumption behavior.

Credamo served solely as the participant recruitment platform. Participants recruited through the platform attended an offline research laboratory in Hengqin, Guangdong, China. An initial group of 310 participants attended the study. Ten cases were excluded due to incomplete sessions, inconsistent response patterns, or invalid questionnaire completion, producing a final sample of 300 participants. The data collection was supported by a small faculty grant from the Chongqing Institute of Engineering, and each participant received the equivalent of USD 2 after completing the study.

The adequacy of the sample size was assessed before the main analysis using G*Power 3.1. For a multiple regression model with a medium effect size of f^2=.15, a significance level of α=.05, and statistical power of.95, the estimated minimum sample size was 160. The final sample of 300 exceeded this requirement under the specified assumptions. The study protocol was reviewed and approved by the Institutional Review Board/Ethics Committee of Shanghai Technical Institute of Electronics and Information (Approval No. STIEI-LLDD-24105). Adult participants provided written informed consent before taking part. For participants younger than 18 years, written consent was obtained from a parent or legal guardian, and the participant also provided assent. Participation was voluntary, and participants could withdraw at any stage without penalty.

Upon arrival at the laboratory, participants were instructed to play their most frequently played and preferred online game for 60 min. Allowing participants to use a familiar game provided a personally relevant context for reporting achievement motivation and flow experience. The duration of play, laboratory conditions, instructions, and post-session survey timing were held consistent across participants. A dual-camera system was used to confirm that participants remained engaged in the gameplay session. Participants completed the questionnaire within one minute after the session, allowing flow experience to be reported close to the gameplay activity.

Achievement Motivation and Flow Experience were assessed using multi-item measures with a 7-point Likert response format ranging from 1 (strongly disagree) to 7 (strongly agree). Achievement Motivation was assessed using a seven-item study-specific measure informed by the achievement dimension of established online gaming motivation frameworks (Yee, 2006; Zhang et al., 2013). The items assessed players' emphasis on advancement, competence, challenge, performance improvement, and successful performance during online gameplay. The measure showed high internal consistency in the current sample (α = .940). Flow Experience was measured with 13 items adapted from the Flow Short Scale developed by Rheinberg et al. (2003) and presented in English by Engeser and Rheinberg (2008). This measure also showed high internal consistency in the current sample (α = .920).

Online Game Consumption Behavior was measured using four ordered monthly expenditure categories: below 100 CNY, 101–500 CNY, 501–1,000 CNY, and above 1,000 CNY. For the regression-based analyses conducted with PROCESS, the four ordered categories were coded from 1 to 4 and treated as a numeric ordered expenditure score, with higher values representing higher monthly expenditure categories. The resulting coefficients therefore indicate associations with movement toward higher expenditure categories and should not be interpreted as changes in exact expenditure amounts. A second factual question recorded monthly top-up frequency using three categories: 1–2 times, 3–5 times, and more than 5 times. Top-up frequency was retained as a supplementary description of participants' purchasing activity and is reported in Table 2. Because these questions recorded factual behavioral categories, Cronbach's alpha, KMO, Bartlett's test, and factor analysis were not applied to Online Game Consumption Behavior.

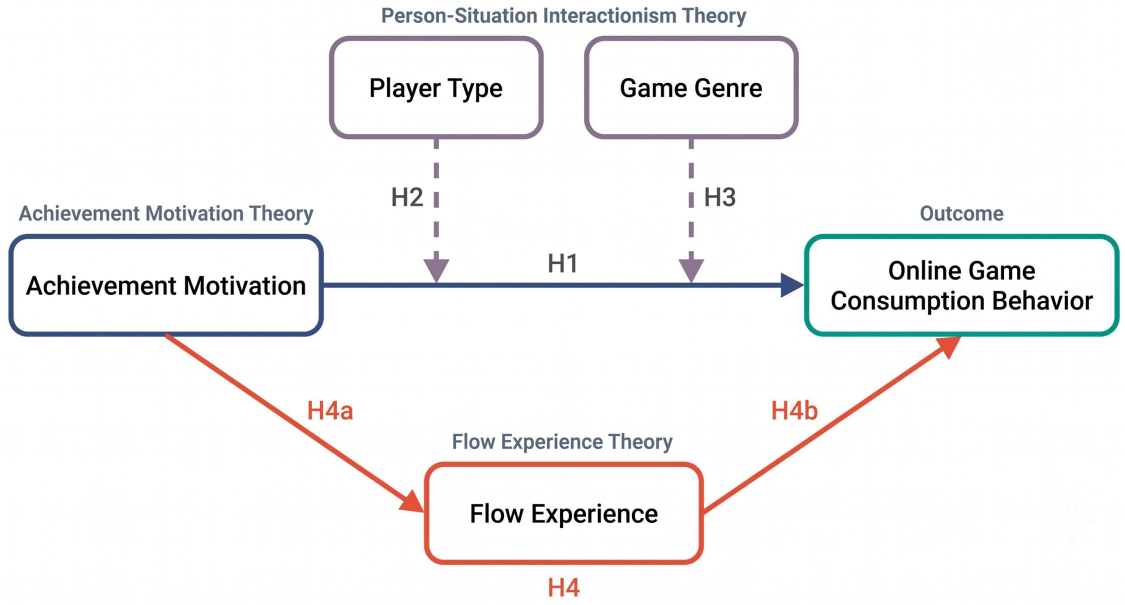
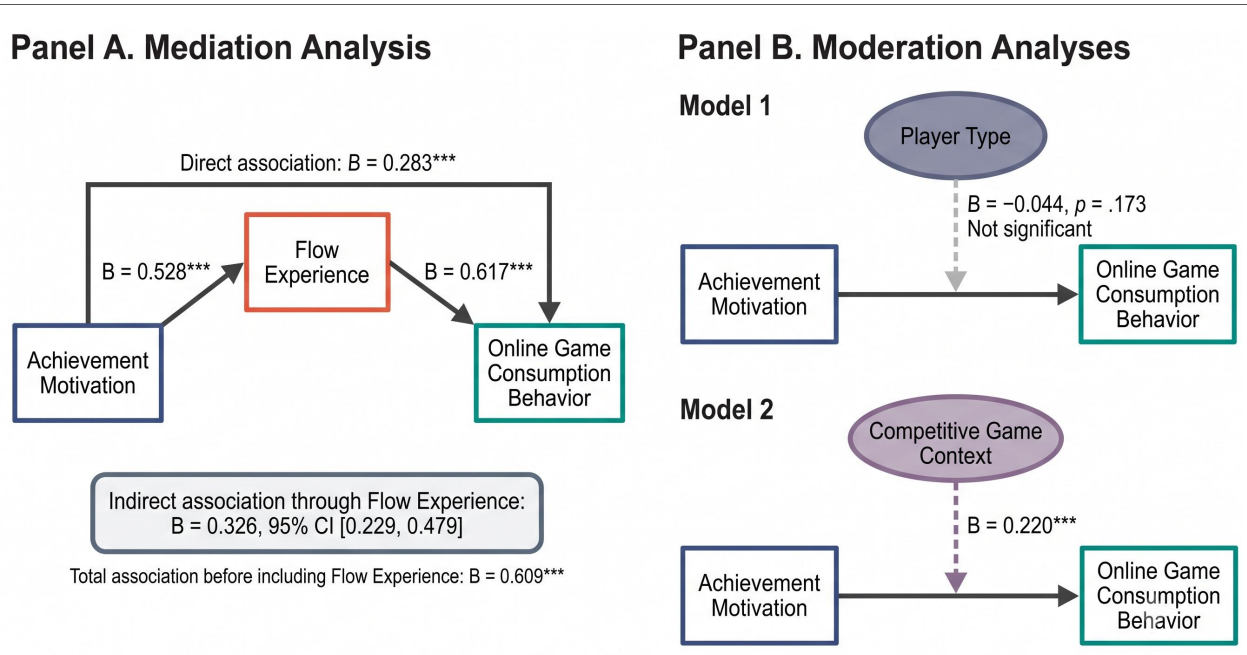


FIGURE 1
Proposed research Model.



Note. Values are unstandardized regression coefficients. The mediation and moderation analyses were estimated separately. *** $p < .001$.

FIGURE 2
Unstandardized coefficients from the mediation and moderation analyses. Panel A presents the mediation analysis, and Panel B presents the two moderation analyses. Values are unstandardized regression coefficients. The mediation and moderation analyses were estimated separately. The confidence interval for the indirect association was generated using 5,000 bootstrap samples. ***$p < .001$.

Player Type was derived from participants' self-reported average weekly gaming hours. Weekly gaming time was initially recorded as a continuous measure and was subsequently coded as a binary moderator. Participants reporting no more than four hours per week were classified as light players, while those reporting more than four hours per week were classified as heavy players. The four-hour cutoff corresponded to the sample distribution and produced 160 light players and 140 heavy players.

Game Genre was recorded from the primary game played during the laboratory session and was coded as a binary moderator according to the dominant competitive structure of the game. MOBA and FPS/TPS games were coded as

TABLE 2 Sample characteristics.

Variable	Category	Frequency (*N*)	Percentage (%)
Gender	Male	153	51.0%
	Female	147	49.0%
Age	Under 18	18	6.0%
	18–25	166	55.3%
	26–35	79	26.3%
	36–45	16	5.3%
	Above 45	21	7.0%
Education Level	Junior High or below	6	2.0%
	High School/ Technical	35	11.7%
	College	68	22.7%
	Undergraduate	161	53.7%
	Master and above	30	10.0%
Occupation	Ordinary Staff	184	61.3%
	Public Institution Staff	47	15.7%
	Service Worker	11	3.7%
	Private Owner/ Entrepreneur	15	5.0%
	Private Enterprise Employee	20	6.7%
	Freelancer	19	6.3%
	Other	4	1.3%
Game Genre	MOBA	103	34.33%
	FPS/TPS	58	19.33%
	RPG	63	21.00%
	Card Games	39	13.00%
	Puzzle Games	37	12.33%
Gaming Platform	Mobile	194	64.67%
	PC	32	10.67%
	Others	74	24.67%
Player Type	≤2 h	108	36.00%
	2–4 h	52	17.33%
	4–6 h	66	22.00%
	6–8 h	45	15.00%
	>8 h	29	9.67%
Monthly In-game Spending	Below 100 CNY	109	36.33%
	101–500 CNY	120	40.00%
	501–1,000 CNY	37	12.33%
	Above 1,000 CNY	34	11.33%
Gaming Experience	Less than 1 year	47	15.67%
	1–3 years	139	46.33%
	4–6 years	58	19.33%
	Above 6 years	56	18.67%
Top-up Frequency per Month	1–2 times	117	39.00%
	3–5 times	95	31.67%
	More than 5 times	88	29.33%

Weekly gaming hours represent the original reported distribution. Player Type represents the binary moderator used in the PROCESS analysis. Online Game Consumption Behavior was represented by the ordered monthly in-game spending categories. Top-up frequency was retained as a supplementary factual indicator of purchasing activity.

competitive contexts due to their direct player-versus-player structure, visible performance comparison, and ranking systems. RPG and puzzle games were coded as noncompetitive contexts when direct competition was not the main form of gameplay. Card games were classified according to the dominant mode of the specific title reported by the participant; titles centered on direct ranked player-versus-player competition were coded as competitive. This coding focused on the competitive structure relevant to H3 while retaining the broader genre categories reported in Table 2.

The data were analyzed using SPSS Statistics 26.0 and PROCESS macro version 4.0 (Hayes, 2018). Descriptive statistics and Pearson correlations were first calculated to summarize the sample and examine the relationships among the principal variables. Internal consistency was assessed only for the multi-item measures of Achievement Motivation and Flow Experience.

H1 was assessed through the association between Achievement Motivation and Online Game Consumption Behavior. The moderating roles of Player Type and Game Genre were examined in separate PROCESS Model 1 analyses. Age and occupation were included as adjustment variables in these two moderation analyses. The indirect association involving Flow Experience was examined using PROCESS Model 4 without additional adjustment variables. Bootstrap confidence intervals were generated from 5,000 resamples.

5 Model analysis and results

5.1 Descriptive statistics and preliminary group comparisons

The initial sample comprised 310 participants recruited through Credamo. Credamo was used only for recruitment, while the gameplay sessions and questionnaire completion took place in the offline laboratory in Hengqin, Guangdong. Data screening identified 10 cases with incomplete sessions, inconsistent response patterns, or invalid questionnaire completion. These cases were excluded, resulting in a final analytical sample of 300 participants and a valid response rate of 96.77%.

The final sample was nearly balanced by gender, with 153 male participants (51.0%) and 147 female participants (49.0%). Participants aged 18–35 accounted for 81.7% of the sample. Undergraduate education was the most frequently reported

educational level (53.7%), and ordinary staff represented the largest occupational group (61.3%). MOBA was the most frequently reported game genre (34.33%), and mobile devices were the most common gaming platform (64.67%). Monthly in-game expenditure below 500 CNY was reported by 76.33% of participants.

One-way analyses of variance were conducted to examine demographic differences in the principal study variables. Achievement Motivation differed significantly across age groups, $F(4, 295) = 21.289$, $p < .001$, as did Online Game Consumption Behavior, $F(4, 295) = 111.057$, $p < .001$, and Flow Experience, $F(4, 295) = 24.750$, $p < .001$. Significant differences across occupational groups were also found for Achievement Motivation, $F(6, 293) = 14.626$, $p < .001$, Online Game Consumption Behavior, $F(6, 293) = 82.893$, $p < .001$, and Flow Experience, $F(6, 293) = 17.804$, $p < .001$. No statistically significant differences were identified across gender or educational groups. Age and occupation were therefore included as adjustment variables in the two moderation analyses examining H2 and H3. Table 2 presents the sample characteristics.

5.2 Measurement assessment and diagnostics

Internal consistency and factorability diagnostics were assessed for the two multi-item measures before the hypothesis tests. Achievement Motivation showed high internal consistency, with a Cronbach's alpha of.940. Its KMO value was.940, and Bartlett's test of sphericity was statistically significant, $\chi^2(21) = 1797.70$, $p < .001$. Flow Experience also showed high internal consistency, with a Cronbach's alpha of.920. Its KMO value was.935, and Bartlett's test of sphericity was statistically significant, $\chi^2(78) = 3268.79$, $p < .001$. These results supported the use of the items within their respective multi-item measures.

Online Game Consumption Behavior was an observed variable represented by a four-level ordered monthly expenditure score. It was therefore not treated as a multi-item scale, and Cronbach's alpha, KMO, Bartlett's test, and factor analysis were not applied to this variable. Pearson correlation analysis showed positive associations between Achievement Motivation and Online Game Consumption Behavior ($r = .629$, $p < .01$), Achievement Motivation and Flow Experience ($r = .636$, $p < .01$), and Flow Experience and Online Game Consumption Behavior ($r = .716$, $p < .01$).

Multicollinearity was also examined for the predictors included together in the outcome model. The VIF values for Achievement Motivation and Flow Experience were both 1.680, indicating no substantial multicollinearity between these predictors. The descriptive statistics, correlations, internal consistency estimates, and factorability diagnostics are presented in Table 3.

5.3 Hypothesis testing

The hypotheses were tested using the PROCESS macro for SPSS. H1 concerned the association between Achievement Motivation and Online Game Consumption Behavior and was examined without additional adjustment variables. The coefficient for Achievement Motivation was positive and statistically significant, $B = .609$, $SE = .044$, $t = 13.983$, $p < .001$, 95% CI [.522,.696]. This result supports H1.

H2 and H3 were examined through separate PROCESS Model 1 analyses, with age and occupation included as adjustment variables in both moderation models. The interaction between Achievement Motivation and Player Type was not statistically significant, $B = -.044$, $SE = .032$, $t = -1.365$, $p = .173$, 95% CI [−.107,.019]. The association between Achievement Motivation and Online Game Consumption Behavior did not differ significantly between light and heavy players; H2 was therefore not supported.

The interaction between Achievement Motivation and Game Genre was statistically significant, $B = .220$, $SE = .059$, $t = 3.717$, $p < .001$, 95% CI [.104,.337]. The positive association between Achievement Motivation and Online Game Consumption Behavior was stronger in competitive game contexts than in noncompetitive game contexts. This result supports H3.

H4 was examined using PROCESS Model 4 without additional adjustment variables. Achievement Motivation was positively associated with Flow Experience, $B = .528$, $SE = .037$, $t = 14.230$, $p < .001$, 95% CI [.455,.601], supporting H4a. Flow Experience was also positively associated with Online Game Consumption Behavior after Achievement Motivation was included in the model, $B = .617$, $SE = .058$, $t = 10.657$, $p < .001$, 95% CI [.503,.731], supporting H4b. The estimated indirect association between Achievement Motivation and Online Game Consumption Behavior through Flow Experience was $B = .326$, with a bootstrapped 95% confidence interval of [.229,.479]. The interval did not include zero. The direct coefficient for Achievement Motivation remained statistically significant after Flow Experience was included, $B = .283$, $SE = .048$, $t = 5.886$, $p < .001$, 95% CI [.188,.378]. This pattern is consistent with partial statistical mediation and supports H4. The results are shown in Table 4. The mediation and moderation estimates are also summarized graphically in Figure 2.

6 Discussion

The positive association between Achievement Motivation and Online Game Consumption Behavior ($B = .609$, $p < .001$) consistent with the view that achievement-oriented players assign greater importance to progress, competence, rank, and successful performance. Costes and Bonnaire (2022), Hamari and Keronen (2017), Lelonek-Kuleta et al. (2021), and Lemmens (2022) similarly connect game-related purchasing with achievement, status, utility, and advancement. Within a commercial game environment, virtual goods may communicate value through their relationship with performance, progression, rarity, or social recognition. Players with stronger Achievement Motivation may consequently view some purchase opportunities as more relevant to their goals. The data do not indicate that every purchase was performance-oriented or that players made fully calculated purchase decisions. They show that higher Achievement Motivation was associated with higher categories of Online Game Consumption Behavior in the present sample.

The mediation results add an experiential component to this relationship. Achievement Motivation was positively associated

TABLE 3 Descriptive statistics, correlations, and measurement diagnostics.

Variable	M	SD	1	2	3	KMO Measure	Bartlett's Test
Achievement Motivation	3.34	0.96	(.940)			.940	$\chi 2$ (21) = 1,797.70, $p < .001$
Flow Experience	3.30	0.80	.636**	(.920)		.935	$\chi 2$ (78) = 3,268.79, $p < .001$
Online Game Consumption Behavior	1.99	0.97	.629**	.716**	-	-	-

$N = 300$. Cronbach's alpha values for the multi-item measures are shown in parentheses on the diagonal. Online Game Consumption Behavior was represented by a single four-level ordered monthly expenditure score. It was not treated as a multi-item scale; therefore, Cronbach's alpha, KMO, and Bartlett's test were not applicable. Monthly top-up frequency was retained as a separate supplementary indicator and was not included in the calculation of Online Game Consumption Behavior.
** $p < .01$.

TABLE 4 Summary of hypothesis test results.

Predictor/Path	B	SE	t	p	95% CI	Hypothesis	Result
Panel A: Total Effect Model							
Dependent Variable: Online Game Consumption Behavior (four-level ordered expenditure score treated as numeric in PROCESS)						H1	Supported
Achievement Motivation (IV)	0.609	0.044	13.983	<.001***	[0.522, 0.696]		
Model Summary: R2 = .396, F(1, 298) = 195.53, $p < .001$							
Panel B: Moderation Analyses							
Dependent Variable: Online Game Consumption Behavior (four-level ordered expenditure score treated as numeric in PROCESS)							
Model 1: Player Type as Moderator						H2	Not Supported
Ach. Motivation×Player Type (Categorical Moderator)	−0.044	0.032	−1.365	.173	[−0.107, 0.019]		
Model Summary: ΔR2 = .004, F(1, 294) = 1.864, $p = .173$							
Model 2: Game Genre as Moderator						H3	Supported
Ach. Motivation×Game Genre (Categorical Moderator)	0.220	0.059	3.717	<.001***	[0.104, 0.337]		
Model Summary: ΔR2 = .027, F(1, 294) = 13.816, $p < .001$							
Panel C: Mediation Analysis							
Step 1: Predicting the Mediator (Path a)						H4a	Supported
Dependent Variable: Flow Experience (Mediator/DV in this step)							
Achievement Motivation (IV)	0.528	0.037	14.230	<.001***	[0.455, 0.601]		
Model Summary: R2 = .405, F(1, 298) = 202.49, $p < .001$							
Step 2: Predicting the Outcome (Paths b and c')						H4b	Supported
Dependent Variable: Online Game Consumption Behavior (four-level ordered expenditure score treated as numeric in PROCESS)							
Achievement Motivation (Latent IV)	0.283	0.048	5.886	<.001***	[0.188, 0.378]		
Flow Experience (Mediator)	0.617	0.058	10.657	<.001***	[0.503, 0.731]		
Model Summary: R2 = .563, F(2, 297) = 191.48, $p < .001$							
Indirect Effect of Achievement Motivation						H4	Supported
Through Flow Experience	0.326	0.064			[0.229, 0.479]		

$N = 300$. B, unstandardized coefficient; SE, standard error; CI, confidence interval. IV, independent variable; DV, dependent variable. Online Game Consumption Behavior was represented by four ordered monthly expenditure categories coded from 1 to 4 and treated as a numeric ordered expenditure score in the PROCESS analyses. The coefficients indicate associations with higher expenditure categories and do not represent changes in exact monetary amounts. Age and occupation were included as adjustment variables only in the two PROCESS Model 1 moderation analyses presented in Panel B. They were not included in the total-effect or mediation models presented in Panels A and C. Panel B reports the interaction terms from two separate moderation analyses. Panel C reports the results of PROCESS Model 4. The confidence interval for the indirect association was generated using 5,000 bootstrap samples and did not include zero.
*** $p < .001$.

with Flow Experience (B = .528, $p < .001$), and Flow Experience remained positively associated with Online Game Consumption Behavior after Achievement Motivation was included in the model (B = .617, $p < .001$). The bootstrapped indirect association was also statistically significant (B = .326, 95% CI [.229,.479]). This pattern is consistent with flow research linking challenge, skill, focused involvement, and immediate feedback with continued engagement in digital games (Cai et al., 2022; Choi

and Kim 2004; Hsu and Lu, 2004; Lin et al., 2025). Chou et al. (2023) also connect flow-related gameplay experiences with in-game purchasing responses.

The mediation finding should be interpreted within the limits of the research design. The result indicates that Flow Experience statistically accounts for part of the association between Achievement Motivation and the categorical consumption outcome.

The significant interaction involving Game Genre (B = .220, $p < .001$) supports the person–situation argument developed in the theoretical framework. Competitive game contexts commonly present visible rankings, direct opposition, performance statistics, and repeated social comparison. These features may increase the relevance of achievement-related goals and make the performance meaning of virtual goods more visible. Brandtner et al. (2022) show that game genres differ in their required skills and player motives, while Jang et al. (2021) and Ma et al. (2021) indicate that genre characteristics can change the relationships between individual characteristics and gaming responses. The present result extends this reasoning to Online Game Consumption Behavior: the positive association with Achievement Motivation was stronger when the dominant game context emphasized competition.

From an Advertising and Marketing Communication perspective, competitive features also form part of the information environment surrounding virtual goods. Rankings, reward comparisons, item attributes, and progression feedback communicate what the game values and how a paid item may relate to player goals. The interaction result nevertheless supports closer attention to the game context in which commercial offers are presented. The meaning of a virtual good may depend partly on whether gameplay makes achievement public, comparable, and repeatedly evaluated.

The nonsignificant interaction involving Player Type (B = -.044, $p = .173$) requires a more limited interpretation than the original manuscript provided. The result does not establish that heavy and light players behave identically, nor does it support the existence of a participation threshold. It shows that the association between Achievement Motivation and Online Game Consumption Behavior did not differ significantly between the two groups in this sample. Weekly gaming hours measure participation duration and exposure to the game environment. They do not directly measure psychological involvement, identification with the game, or personal importance. The result therefore cautions against treating time-based categories as complete measures of player involvement.

6.1 Theoretical implications

The first theoretical contribution concerns the position of Online Game Consumption Behavior within Advertising and Marketing Communication. Research on commercial communication in games has commonly examined brand placement, in-game advertising, and advergames (Nelson, 2002; Terlutter and Capella, 2013). These formats generally concern commercial messages presented through game content. The present study draws attention to communication occurring through the game's own monetization system. Item descriptions, rarity indicators, rankings, reward systems, and progress information communicate the functional and symbolic value of virtual goods. Online Game Consumption Behavior can therefore be considered in relation to the commercial meanings produced through gameplay and the purchase interface.

This contribution does not imply that game mechanics are identical to conventional advertisements. It shows how marketing communication can operate across the platform experience through repeated information about goals, rewards, status, and available virtual goods. Hofacker et al. (2016) similarly explain how game design principles can influence consumer engagement in digital marketing settings. The current results connect this communication environment with variation in player motivation and categorical consumption behavior.

The second contribution concerns the integration of Achievement Motivation and Flow Experience. Achievement Motivation explains why mastery, challenge, feedback, and visible performance may hold different levels of personal relevance across players. Flow Experience represents the reported quality of focused involvement during gameplay. Their statistical relationship with Online Game Consumption Behavior shows that motivational orientation and gameplay experience provide related but distinct information. The significant indirect association supports the inclusion of Flow Experience in the model without redefining it as a new form of consumption or assuming that flow necessarily produces purchasing.

This interpretation also adds caution to the relationship between enjoyment and commercial behavior. Hamari (2015) shows that enjoyment, continued use, attitudes toward virtual goods, and purchasing do not follow an identical pattern. The present findings are consistent with this distinction. Flow Experience was associated with Online Game Consumption Behavior, but the study did not measure the specific reasons behind each purchase. Flow should therefore remain an experiential construct, while Online Game Consumption Behavior remains a separate observed categorical outcome.

The third contribution concerns the role of Game Genre. Person–situation interactionism proposes that individual characteristics acquire different behavioral relevance under different situational conditions. The significant Game Genre interaction is consistent with this position. Achievement Motivation showed a stronger association with Online Game Consumption Behavior in genres coded as competitive, where performance comparison and ranking are more visible. The result supports the inclusion of Game Genre as a moderator and indicates that consumer responses to monetization cannot be explained by player characteristics alone.

For communication research, this contextual result suggests that the interpretation of a commercial offer depends on the information structure surrounding it. A virtual item associated with progress or performance may carry a different meaning in a ranked competitive game than in a game centered on narrative, exploration, or individual problem solving. The contribution lies in connecting the player's motivational orientation with the context through which virtual-item value is communicated. It does not require a new theoretical label beyond person–situation interactionism.

The fourth contribution arises from the nonsignificant Player Type interaction. Weekly gaming time is useful for describing participation, but the result does not support its use as a

complete indicator of psychological involvement. This distinction improves conceptual consistency. Player Type in the current study reflects duration of play, while involvement would require direct measures of personal relevance, emotional attachment, or identification.

6.2 Practical implications

The findings provide several cautious implications for the communication of virtual goods. Achievement-oriented players may assign greater relevance to offers connected with progress, competence, rank, or visible accomplishment. Game publishers can communicate these functions clearly through accurate item descriptions and transparent explanations of how a virtual good relates to gameplay. Functional advantages, cosmetic features, purchase duration, probability-based rewards, and usage restrictions should be presented in a form that players can understand before making a purchase.

The significant Game Genre interaction indicates that competitive context deserves particular attention. Commercial messages presented alongside rankings, performance gaps, or win–loss feedback may carry greater relevance for achievement-oriented players. This increased relevance also creates a responsibility to avoid misleading claims or excessive pressure. Paid items should not be presented as necessary for personal worth, social acceptance, or legitimate participation. Clear distinctions between skill-based achievement and paid advantages can support player trust and reduce concerns surrounding unfair monetization.

The results involving Flow Experience do not justify targeting players at moments of frustration or presenting purchases as necessary to continue an enjoyable experience. A more responsible application is to design commercial interfaces that do not unnecessarily interrupt gameplay and that give players sufficient information and time to evaluate an offer. Optional convenience items and additional content can be communicated without concealing costs or creating artificial barriers designed solely to prompt payment. This approach is especially important when underage players may encounter the commercial environment.

The nonsignificant Player Type interaction suggests that weekly gaming hours should not be the sole basis for player segmentation. Time-based categories describe usage frequency but provide limited information about the goals a player values. Managers may obtain a clearer understanding through voluntary preference settings, purchase histories, preferred gameplay modes, and direct feedback about player goals. Any personalized communication should follow appropriate privacy standards and allow players to manage promotional settings.

More broadly, the findings support coordination between game design and marketing communication. Rankings, reward systems, item descriptions, storefronts, and purchase prompts contribute jointly to how players understand virtual goods. Consistency across these elements can reduce confusion about product value and expected gameplay outcomes. Transparent communication is especially important in competitive environments, where achievement-related messages may be more strongly associated with Online Game Consumption Behavior.

7 Limitations and future research

The study has several limitations that define the scope of its findings. First, the research used a laboratory-based cross-sectional correlational design. Achievement Motivation, Flow Experience, and Online Game Consumption Behavior were measured within one data-collection session, and the study did not manipulate the focal variables or randomly assign participants to conditions. The results therefore identify direct, indirect, and conditional statistical associations but do not establish causal effects or the temporal order among the variables. Longitudinal studies could measure motivation, gameplay experience, and consumption at multiple points in time. Experimental studies could also manipulate specific competitive or commercial communication features, such as ranking visibility, reward descriptions, item presentation, or performance feedback, to examine how players respond to these features under controlled conditions.

Second, the laboratory procedure provided consistent session duration, instructions, and survey timing, but it also limits external validity. Participants played their preferred games in an offline laboratory, which differs from gameplay at home or in other natural settings. Laboratory monitoring may have influenced concentration, gameplay behavior, and questionnaire responses. Participants also played different titles across several genres and platforms. This approach increased personal familiarity with the game being evaluated, but it reduced control over differences in game mechanics, monetization systems, difficulty, and commercial content. Field studies conducted during players' normal gaming routines would provide useful evidence concerning the extent to which the present relationships apply outside the laboratory.

Third, the study relied on self-reported measures. Achievement Motivation and Flow Experience were assessed using multi-item questionnaires, while Online Game Consumption Behavior was represented by ordered categories of reported monthly expenditure. Self-reported expenditure may be affected by memory errors, social desirability, or unwillingness to disclose purchasing behavior. Treating the four ordered categories as a numeric expenditure score also means that the coefficients should be interpreted as approximate movement toward higher expenditure categories, rather than as changes in exact monetary spending. The categorical measure also does not capture exact expenditure amounts, individual transactions, refunds, gifts, or changes in spending over time. Future research could combine survey measures with anonymized transaction records, account-level purchase histories, or spending diaries. Such data would allow researchers to compare reported consumption with observed purchasing activity.

The measurement assessment also has limits. Internal consistency was examined for Achievement Motivation and Flow Experience using Cronbach's alpha, and KMO and Bartlett's tests were reported as factorability diagnostics. Future research should assess the measurement structure using CFA and report standardized factor loadings, composite reliability, average variance extracted, model-fit indices, and discriminant-validity evidence. Collecting predictor and outcome information at different times or from different data sources would also reduce concerns associated with common method variance.

Fourth, the sample was recruited through Credamo and completed the study in an offline laboratory in Hengqin,

Guangdong. The participants were predominantly young adults from Mainland China, with a high proportion reporting undergraduate education. Cultural expectations concerning competition, status, digital payments, and virtual goods may differ across countries and player communities. Replication with participants from different cultural, age, educational, and socioeconomic groups would clarify the stability of the proposed relationships. Cross-national research could also examine whether achievement-related marketing communication carries similar meaning across gaming markets.

Fifth, Player Type was operationalized by dividing weekly gaming hours into light and heavy player categories using a four-hour cutoff. This binary measure provides a clear comparison but reduces the variation contained in the original weekly-hours measure. Players within the same category may differ considerably in experience, psychological involvement, preferred gameplay modes, and personal attachment to the game. The nonsignificant H2 result applies only to this time-based operationalization and does not show that player involvement is unimportant. Future research should analyze weekly gaming time as a continuous moderator and include validated measures of personal relevance, identification, commitment, and social participation.

Finally, Game Genre was coded according to the dominant competitive structure of the game. This binary distinction supported the test of H3, but it simplifies the diversity within individual genres and titles. RPG and card games, for example, may include competitive, cooperative, and individual modes within the same title. Players may also move between these modes during a single gaming session. Future studies could code competitive context at the level of the specific game mode, ranking system, opponent structure, or commercial feature encountered by the player.

8 Conclusion

This study examined the relationships among Achievement Motivation, Flow Experience, Game Genre, Player Type, and Online Game Consumption Behavior. Data were collected from 300 online game players through a controlled, laboratory-based cross-sectional correlational study. The proposed model connected achievement motivation theory, flow theory, and person–situation interactionism to explain variation in categorical game consumption behavior.

Achievement Motivation was positively associated with Online Game Consumption Behavior. Flow Experience accounted for part of this association, while the direct association remained statistically significant. The relationship between Achievement Motivation and Online Game Consumption Behavior was stronger in competitive game contexts. Weekly gaming time, classified into light and heavy Player Type, did not significantly change this relationship. These results support H1, H3, and H4, while H2 was not supported.

The study contributes to Advertising and Marketing Communication by showing how online game consumption can be considered within a commercial environment formed through gameplay and purchase-related information. Rankings, rewards, item descriptions, progression systems, and competitive feedback communicate the functional and symbolic value of virtual goods. The findings suggest that player motivation and gameplay experience are relevant to how players respond to this commercial environment, while Game Genre was associated with variation in the strength of the observed relationship.

The results also offer cautious guidance for game publishers and marketers. Communication surrounding virtual goods should clearly explain their functions, prices, restrictions, and relationship with gameplay. Competitive contexts require particular attention to transparency, fairness, and the protection of younger or potentially vulnerable players. Weekly gaming time alone provides limited information for understanding player responses and should not serve as the sole basis for segmentation or personalized offers.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Ethics Statement

The study protocol was reviewed and approved by the Institutional Review Board/Ethics Committee of Shanghai Technical Institute of Electronics and Information (Approval No. STIEI-LLDD-24105). Adult participants provided written informed consent before taking part. For participants younger than 18 years, written consent was obtained from a parent or legal guardian, and the participant also provided assent.

Author contributions

YH: Conceptualization, Data curation, Formal analysis, Investigation, Resources, Software, Validation, Visualization, Writing – original draft. JG: Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

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